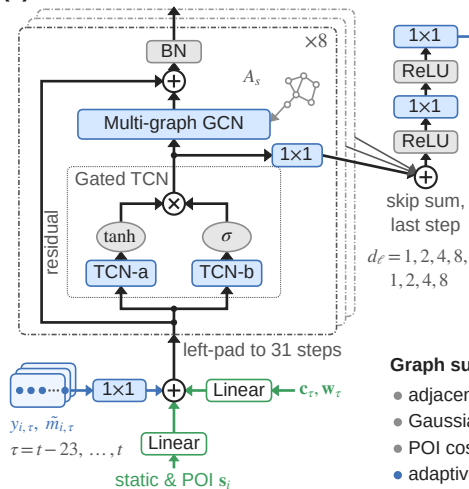
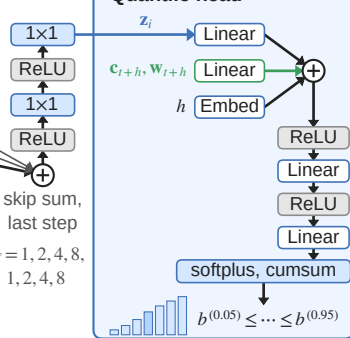


(a) Price-blind ST backbone

Graph supports A_s (2-hop diffusion)

- adjacency (symmetrised)
- Gaussian distance kernel
- POI cosine similarity (top 10)
- adaptive $\tilde{A} = \text{softmax}(\text{ReLU}(E_1 E_2^T))$

Quantile head



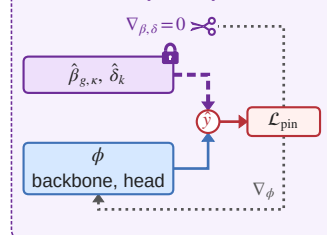
(b) Anchored price response

$\kappa(t+h)$	Workday		Rest day		
$g(i)$	day	night	day	night	
Low-density	○	○	○	○	
Residential	○	●	●	●	
Commercial	○	●	○	●	

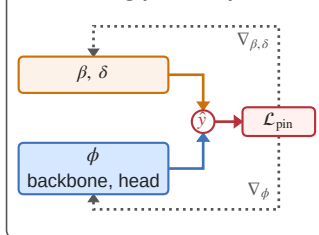
(illustrative)

$$\eta = \hat{\beta}_{g(i), \kappa(t+h)} \Delta \ell + \Sigma_k \hat{\delta}_k S^{(k)}$$

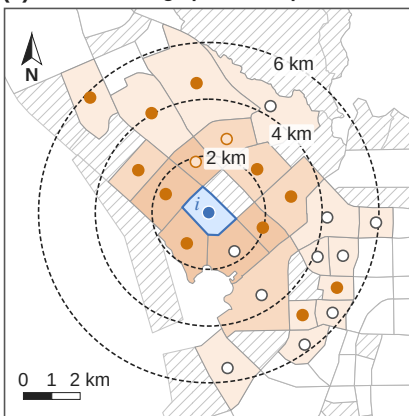
Cut feedback (default)



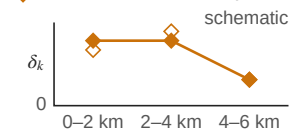
Joint training (ablations)



(c) Distance-ring spillover exposure



- zone i (TOU, commercial)
- ring 1: [0, 2) km, 5 zones
- ring 2: [2, 4) km, 5 zones
- ring 3: [4, 6) km, 13 zones
- zones beyond 6 km
- no charging data
- strict TOU
- weak TOU
- fixed price ($\Delta \ell_j \equiv 0$)
- raw $\hat{\delta}_k$ from switch events
- used: PAV to $\delta_1 \geq \delta_2 \geq \delta_3 \geq 0$



$$S_{i,t}^{(k)} = \text{mean}\{\Delta \ell_{i,t} : j \in \mathcal{R}_k(i)\}, \quad \mathcal{R}_k(i) = \{j \neq i : r_{k-1} \leq d_{ij} < r_k\}, \quad d_{ij}: \text{centroid distance}$$

$$\eta_{i,t}^{\text{spill}} = \Sigma_k \delta_k S_{i,t}^{(k)}, \quad \text{an empty ring gives } S_{i,t}^{(k)} = 0$$

(d) Masked adaptive conformal calibration

